

A New Reconstruction Formula and Shift Frame for WGFRFT

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ABSTRACT

The study of signals defined on irregular domains has garnered significant attention, and the field of graph signal processing has emerged as a way to model such signals using underlying graph structures. Vertex frequency analysis is a crucial tool for graph signal analysis and representation. Finding different kinds of frames via the windowed graph fractional Fourier transform (WGFRFT) is necessary to extract the characteristics of graph signals. To overcome this issue, we introduce the concept of the dual of WGFRFT frame and use it to develop a new reconstruction formula for graph signals. Additionally, we propose fractional shift frame. Finally, to highlight the utility of our proposed frames, we present some examples which use our fractional shift frame to extract the spectrum of graph signal.

Keywords: Graph signal processing, Graph Fourier transform, Fractional Fourier transform, Frame, Vertex frequency analysis.

1. INTRODUCTION

Signals characterized by high dimensions, discreteness, and irregularity, are gaining importance [1–4]. Graphs have emerged as a crucial way to represent such signals as they can describe their geometric and topological structures [5, 6]. In recent years, some key signal processing methods have been extended to graph signals, resulting in the development of graph-based transforms such as the graph Fourier transform (GFT) [3, 7–10], graph filters [11, 12], sampling on graphs [1, 6, 13–16], uncertainty principles [17, 18]. These advancements have been applied in various domains, including social and sensor networks, machine learning, and deep neural networks.

In graph signal processing (GSP), the vertex-frequency analysis is a hot issue, which is similar to the time-frequency analysis in classical signal processing. This analysis involves various graph-based transforms and the associated frame theory [2, 19–26]. A new framework for constructing Gabor-type frames with sharp bounds for graph signals has been proposed [21]. The classical frame can be written in terms of LocLets which is the wavelets localized in the Fourier domain [23]. Several local-set-based frames are introduced for constructing reconstruction methods [24]. By means of translation on the graph, a spectrum-adapted compact point frequency framework is proposed [25]. Signal-adapted tight frames on graphs are designed and compared with non-signal-adapted frames [26]. Among them, the windowed graph Fourier transform (WGFT) can indicate characteristics of graph signals in the vertex-frequency domain [2, 27]. Various properties of the WGFT and related frames have been explored.

In order to better display the vertex frequency information of graph signals, fractional order as a new research direction has been introduced in graph signal processing (GSP) [1, 9, 10, 12, 20, 28–32]. Like it in graph Fourier domain, window graph fractional Fourier transform and associated frame have been proposed in our previous paper [20]. However, in [20], we only show a simple frame of the window function. Frame theory is complex, so more content needs to be produced to supply the frame theory based on WGFRFT [33]. In this paper, with the definition of WGFRFT, we study the duals of WGFRFT frames to establish a new reconstruction formula for any graph signals. Then, equipped with the fractional shift operator, we explore the fractional shift frame in graph fractional domain to further broaden the scope of vertex-frequency analysis. Finally, we use the synthetic data to demonstrate that our fractional shift frame effectively shows vertex-frequency characteristics of graph signals.

2. Preliminaries

In this section, we first introduce the frame and operators. Then the definition of WGFRFT is given to lead out the concept about the frame of the window function.

2.1 Frame and operators [33]

A frame is a collection of vectors that span the signal space and provide a redundant representation of signals. This redundancy can be used to facilitate signal reconstruction [21].

Definition 1. A frame can be represented mathematically as a collection of vectors $\{f_k\}_{k=1}^M \subseteq \mathbb{R}^N$ ($M \geq N$) in a Hilbert space H . $\{f_k\}_{k=1}^M \subseteq \mathbb{R}^N$ ($M \geq N$) form a finite frame, if there are constants $A, B > 0$ make

$$A \|f\|^2 \leq \sum_{k=1}^M |\langle f, f_k \rangle|^2 \leq B \|f\|^2$$

hold for $f \in \mathbb{R}^N$. Then A and B are bounds of frame.

The analysis operator T^* is a linear operator that takes a signal and decomposes it into a set of coefficients with respect to a given frame. The synthesis operator T is a linear operator that takes a set of coefficients and synthesize a signal in the frame.

1. Analysis operator:

$$T^* f = \{\langle f, f_k \rangle\}$$

2. Synthesis operator:

$$T(\{c_k\}) = \sum_{k=1}^M c_k f_k$$

Together, the analysis and synthesis operators form a frame decomposition of a signal. They form a set of dual operators. Let $S = TT^*$ represent the frame operator of $\{f_k\}$, i.e.

$$Sf = \sum_{k=1}^M \langle f, f_k \rangle f_k, f \in \mathbb{R}^N$$

If there exists another set of vectors $\{g_k\}_{k=1}^M$ satisfying

$$f = \sum_{k=1}^M \langle f, g_k \rangle f_k, f \in \mathbb{R}^N$$

then $\{g_k\}$ is a dual frame of $\{f_k\}$. A dual frame is a set of vectors that allows the synthesis operator to be expressed as a linear combination of the analysis operator. This makes it possible to reconstruct the original signal from its coefficients using the dual frame.

2.2 Windowed graph Fractional Fourier Transform

Given an undirected graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{W}\}$ which consists of a finite set of vertices $\mathcal{V} = \{v_0, \dots, v_{N-1}\}$, a set of edges $\mathcal{E} = \{(i, j) | i, j \in \mathcal{V}, j \sim i\} \subseteq \mathcal{V} \times \mathcal{V}$. $N = |\mathcal{V}|$ is the number of vertices and $\mathcal{W}$ is the adjacency matrix. $\mathcal{W} = [\mathcal{W}_{ij}] \in \mathbb{R}^{N \times N}$ is defined as $\mathcal{W}_{ij} = 1$ if $(i, j) \in \mathcal{E}$ and $\mathcal{W}_{ij} = 0$ otherwise. The graph Laplacian is defined as $\mathcal{L} = D - \mathcal{W}$, where $D := \text{diag}(d_1, \dots, d_N)$ is a diagonal degree matrix of $\mathcal{G}$. Suppose that the Laplacian matrix has eigendecomposition as $\mathcal{L} = \chi \Lambda \chi^H$, where $\{\chi_0, \chi_1, \dots, \chi_{N-1}\}$ is the set of orthonormal eigenvectors, and $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1} := \lambda_{\max}$ is the corresponding eigenvalues. These eigenvalues form a diagonal matrix $\Lambda = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{N-1}])$. The superscript H represents the Hermitian transpose operation.

Then we give the definition of the graph fractional Laplacian operator $\mathcal{L}_\alpha = \kappa R \kappa^H$, where α is the fractional order, $0 < \alpha \leq 1$, $\ell = 0, 1, \dots, N-1$ [10]. And

$$\begin{aligned} \kappa &= [\kappa_0, \kappa_1, \dots, \kappa_{N-1}] = \chi^\alpha \\ R &= \text{diag}([r_0, r_1, \dots, r_{N-1}]) = \Lambda^\alpha \end{aligned}$$

where $r_\ell = \lambda_\ell^\alpha$. Using $\{\kappa_i\}$, the graph fractional translation operator $(T_i^\alpha f)(n)$ and graph fractional modulation operator $(M_k^\alpha f)(n)$ are defined as

$$(T_i^\alpha f)(n) = (\sqrt{N})^\alpha \sum_{\ell=0}^{N-1} \hat{f}_\alpha(\ell) \kappa_\ell^*(i) \kappa_\ell(n)$$

and

$$(M_k^\alpha f)(n) = (\sqrt{N})^\alpha f(n) \kappa_k(n)$$

Let

$$\Psi_{i,\ell}(n) = (M_\ell^\alpha T_i^\alpha \psi)(n) = N^\alpha \kappa_\ell(n) \sum_{m=0}^{N-1} \hat{\psi}_\alpha(m) \kappa_m^*(i) \kappa_m(n)$$

and $\psi \in L^2(\mathbb{R})$ is a window function, $\hat{f}_\alpha(\ell)$ represents the spectral graph fractional Fourier transform for f . The windowed graph fractional Fourier transform (WGFRFT) $(W_{\psi,\alpha} f)(i, \ell)$ of a graph signal f is

$$(W_{\psi,\alpha} f)(i, \ell) = \langle f, \Psi_{i,\ell} \rangle = N^\alpha \sum_{d=1}^N f(d) \left(\sum_{m=0}^{N-1} \hat{\psi}_\alpha^*(m) \kappa_m(i) \kappa_m^*(d) \right) \kappa_i^*(d).$$

2.3 A property of the window function

Theorem 1. *If a window function $\psi \in L^2(\mathbb{R})$ satisfies $T_i^\alpha \psi \neq 0$, then the family $\{\Psi_{i,\ell}\}_{i=1,2,\dots,N; \ell=0,1,\dots,N-1}$ forms a frame, thus for any signal $f \in \mathbb{R}^N$*

$$0 < A \|f\|_2^2 \leq \sum_{i=1}^N \sum_{\ell=0}^{N-1} |\langle f, \Psi_{i,\ell} \rangle|^2 \leq B \|f\|_2^2 < \infty,$$

where

$$A = \min_{n \in \{1,2,\dots,N\}} \{N^\alpha \|T_n^\alpha \psi\|_2^2\}$$

and

$$B = \max_{n \in \{1,2,\dots,N\}} \{N^\alpha \|T_n^\alpha \psi\|_2^2\}.$$

3. Dual and shift frame of WGFRFT

3.1 Dual frame

Graph signal reconstruction is closely related to frame theory. For any graph signal $f \in L^2(\mathbb{R})$, it can be reconstructed by the WGFRFT coefficients and a dual frame. First, we prove that the frame $\Psi_{i,l}$ has a dual $\tilde{\Psi}_{i,l}$

Theorem 2 (WGFRFT Frame and Dual). *Let $\Psi_{i,l}$ be a WGFRFT frame defined in Theorem 1. For any two sets of functions ψ and $\tilde{\psi}$, if there exists a constant μ such that $\hat{\psi}_\alpha(m) \hat{\tilde{\psi}}_\alpha(m) = \mu$, we say that $\tilde{\Psi}_{i,l}$ is a dual of $\Psi_{i,l}$. Then, the following hold:*

For any signal f , we have $f = C \sum_{l=0}^{N-1} \sum_{i=1}^N \langle f, \tilde{\Psi}_{i,l} \rangle \Psi_{i,l} = C \sum_{l=0}^{N-1} \sum_{i=1}^N \langle f, \Psi_{i,l} \rangle \tilde{\Psi}_{i,l}$, where C is a constant.

Proof.

$$\begin{aligned}
& \sum_{i=1}^N \sum_{l=0}^{N-1} \langle f, \Psi_{i,l} \rangle \tilde{\Psi}_{i,l}(n) \\
&= \sum_{i=1}^N \sum_{l=0}^{N-1} \left(N^\alpha \sum_{q=1}^N f(q) \kappa_l^*(q) \sum_{m=0}^{N-1} \hat{\psi}_\alpha(m) \kappa_m(i) \kappa_m^*(q) \right) \\
&\quad \left(N^\alpha \kappa_l(n) \sum_{m'=0}^{N-1} \hat{\psi}_\alpha(m') \kappa_{m'}^*(i) \kappa_{m'}(n) \right) \\
&= (N^\alpha)^2 \sum_{q=1}^N f(q) \sum_{m=0}^{N-1} \sum_{m'=0}^{N-1} \hat{\psi}_\alpha(m) \hat{\psi}_\alpha(m') \kappa_m^*(q) \kappa_{m'}(n) \\
&\quad \sum_{i=1}^N \kappa_m(i) \kappa_{m'}^*(i) \sum_{l=0}^{N-1} \kappa_l^*(q) \kappa_l(n) \\
&= (N^\alpha)^2 \sum_{q=1}^N f(q) \sum_{m=0}^{N-1} \sum_{m'=0}^{N-1} \hat{\psi}_\alpha(m) \hat{\psi}_\alpha(m') \kappa_m^*(q) \kappa_{m'}(n) \delta_{mm'} \delta_{qn} \\
&= (N^\alpha)^2 f(n) \sum_{m=0}^{N-1} \hat{\psi}_\alpha(m) \hat{\psi}_\alpha(m) |\kappa_m(n)|^2 \\
&= (N^\alpha)^2 \mu f(n)
\end{aligned}$$

Note that the dual $\tilde{\Psi}_{i,l}$ is also a WGFRT frame. The proof is similar to Theorem 1 in [20].

Theorem 3 (Reconstruction Formula). *From the above Proof, it's easy to get that the signal f can be reconstructed by*

$$f(n) = \frac{\sum_{i=1}^N \sum_{l=0}^{N-1} \langle f, \Psi_{i,l} \rangle \tilde{\Psi}_{i,l}(n)}{(N^\alpha)^2 \mu}$$

3.2 Fractional shift frame

By utilizing the graph shift operator as a translation operator to substitute $(T_i^\alpha f)(n)$ in frame $\Psi_{i,\ell}(n)$, it becomes achievable to reduce the complexity of the frame [27]. In GSP, the adjacency matrix acts as a shift operator [34]. In graph fractional domain, the fractional adjacency matrix S_F can also be used this way. Assume that the eigendecomposition of the graph adjacency matrix A is

$$A = VUV^{-1}$$

then the fractional shift operator [1] is

$$S_F = (\tilde{s}_0, \tilde{s}_1, \dots, \tilde{s}_{N-1}) = V^\alpha U (V^{-1})^\alpha$$

The definition of fractional shift frame operator also conforms to the process of translation and modulation of window function.

Theorem 4 (Fractional shift frame). *Given a window function $\psi \in \mathbb{R}^N$, with $S_F \psi \neq 0$. Define $\Psi_i = \kappa_i \circ (S_F \psi)$ where $\circ$ denotes Hadamard product. $\{\Psi_i\}$ forms a frame when $\tilde{s}_j \psi \psi^* \tilde{s}_j^* > 0$ for all j . The optimal lower and upper frame bounds are $A = \min_{j \in \{1,2,\dots,N\}} \tilde{s}_j \psi \psi^* \tilde{s}_j^*$ and $B = \max_{j \in \{1,2,\dots,N\}} \tilde{s}_j \psi \psi^* \tilde{s}_j^*$*

Proof. In this proof, we mainly apply the definition in Section 2.1 $S = TT^*$ to construct the fractional shift frame. First, we use matrix $\Psi_i = D_i S_F \psi$ to represent $\Psi_i = \kappa_i \circ (S_F \psi)$, where $D_i = \text{diag}(\kappa_{i,0}, \kappa_{i,1}, \dots, \kappa_{i,N-1})$. The synthesis operator of Ψ_i is

$$T = (D_1 S_F \psi, D_2 S_F \psi \dots, D_N S_F \psi)$$

According to Section 2.1, the frame operator of $\{\Psi_i\}$ can be written as

$$\begin{aligned}
TT^* &= \sum_{i=1}^N D_i S_F \psi \psi^* S_F^* D_i^* \\
&= [(S_F \psi)(S_F \psi)^*] \circ \left(\sum_{i=1}^N \kappa_i \kappa_i^* \right) \\
&= [(S_F \psi)(S_F \psi)^*] \circ I_N
\end{aligned}$$

where I_N is the $N \times N$ identity matrix.

The diagonal entries of TT^* are given by the diagonal entries of $S_F \psi \psi^* S_F^*$. Using simple matrix computation, we can express the j th diagonal entry of $S_F \psi \psi^* S_F^*$ as $\tilde{s}_j \psi \psi^* \tilde{s}_j^*$. The the frame bounds of $\{\Psi_i\}$ correspond to the minimum and maximum values of $\tilde{s}_j \psi \psi^* \tilde{s}_j^*$.

4. Application

In this section, to assess the effectiveness of the proposed fractional shift frame (FSF) in extracting vertex frequency characteristics from graph signals, we conduct a simulation experiment based on toolbox in [35]. A path graph with 128 vertices is considered as the underlying structure. We adopt the $\kappa_{64}(n)$ from the graph fractional Laplacian operator of this path graph as a signal. The second signal we use is the summation of $\kappa_{42}(n)$, $\kappa_{83}(n)$, and $\kappa_{128}(n)$, and restrict them on the first 42, 43-83, 84-128 vertices respectively. The fractional order is $\alpha=0.9$. The results in Fig. 1 show that our method captures the frequency correctly.

5. Conclusions

In conclusion, the field of graph signal processing has gained significant attention in recent years due to the need to analyze signals defined on irregular domains. In this article, the dual of the WGFRFT frame and a reconstruction formula are elaborated, then the fractional shift frame is introduced. The proposed frames are demonstrated to be useful through various examples that employed the fractional shift frames to extract the spectrum of graph signals. Our work offers valuable insights to the development of the field of graph signal processing, enabling researchers to better analyze and represent signals on irregular domains.

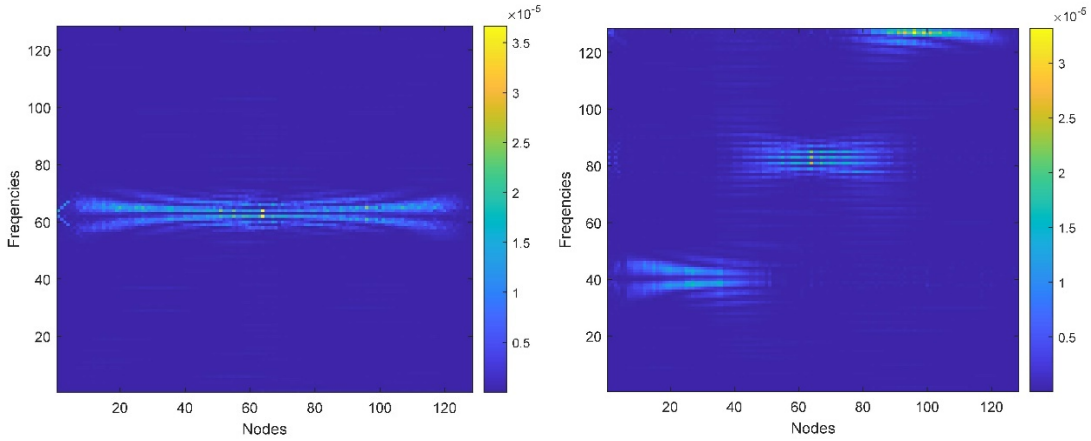


Figure 1. Vertex frequency representations of signals using the fractional shift frame with fractional order $\alpha=0.9$.

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